COMP9602: Optimization

Duality (I)

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Roadmap

Theory	convex set convex function standard forms of optimization problems, quasi-convex optimization linear program, integer linear program quadratic program geometric program semidefinite program vector optimization duality
Algorithm	unconstrained optimization equality constrained optimization interior-point method localization methods subgradient method decomposition methods

Duality

- Optimization in standard form (not necessarily convex)

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

D=dom $f_0 \cap$ dom $f_1 \cap \ldots \cap$ dom f_m $\cap$ dom $h_1 \cap \ldots \cap$ dom h_p

variable $x \in \mathbf{R}^n$, domain $\mathcal{D}$, optimal value $p^\star$

- Lagrangian: $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$, with $\mathbf{dom}\, L = \mathcal{D} \times \mathbf{R}^m \times \mathbf{R}^p$

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x)$$

- Lagrange multipliers (Lagrangian dual variables):

  λ_i is Lagrange multiplier associated with $f_i(x) \leq 0, i = 1, \ldots, m$

  ν_i is Lagrange multiplier associated with $h_i(x) = 0, i = 1, \ldots, p$

- augment $f_0(x)$ by weighted sum of constraint functions (penalty functions)

Lagrange dual

Lagrange dual function: $g : \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$,

$$\begin{aligned} g(\lambda, \nu) &= \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \\ &= \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \right) \end{aligned}$$

- Lagrange dual problem

$$\begin{aligned} &\max g(\lambda, \nu) \\ \text{subject to:} \quad &\lambda \succeq 0 \end{aligned}$$

Note that there is no positivity constraint on ν

LP and its dual

- Standard form LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \succeq 0 \end{array}$$

- Lagrangian:

$$\begin{aligned} L(x, \lambda, \nu) &= c^T x + \nu^T (Ax - b) - \lambda^T x \\ &= -b^T \nu + (c + A^T \nu - \lambda)^T x \end{aligned}$$

- Lagrange dual function

$$g(\lambda, \nu) = \inf_{x \in R^n} L(x, \lambda, \nu) = \begin{cases} -b^T \nu & A^T \nu - \lambda + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

- Dual problem

$$\begin{array}{ll} \text{maximize} & -b^T \nu \\ \text{subject to} & A^T \nu + c \succeq 0 \end{array}$$

Lagrange dual and conjugate function

Optimization with affine inequality and equality constraints

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & Ax \preceq b, \quad Cx = d \end{array}$$

Lagrange dual function

$$\begin{aligned} g(\lambda,\nu) &= \inf_{x \in \mathbf{dom}\, f_0} \left(f_0(x) + (A^T\lambda + C^T\nu)^T x - b^T\lambda - d^T\nu\right) \\ &= -f_0^*(-A^T\lambda - C^T\nu) - b^T\lambda - d^T\nu \end{aligned}$$

- recall definition of conjugate $f^*(y) = \sup_{x \in \mathbf{dom}\, f}(y^T x - f(x))$

(Slide 28, Convex Function notes)

- simplifies derivation of dual if conjugate of f_0 is known

$$f_0(x) = \sum_{i=1}^n x_i \log x_i, \qquad f_0^*(y) = \sum_{i=1}^n e^{y_i - 1}$$

Example on deriving the dual problem

$$\begin{aligned} \min \quad & x_1^2 + x_2^2 \\ \text{s.t.} \quad & 3x_1 + 5x_2 = 7 \\ & 6x_1 + 4x_2 \leq 9 \end{aligned}$$

Lagrangian:

$$L(x, \lambda, \mu) = x_1^2 + x_2^2 + \lambda(6x_1 + 4x_2 - 9) + \mu(3x_1 + 5x_2 - 7)$$

Lagrange dual function:

$$\begin{aligned} g(\lambda, \mu) &= \inf_{x \in R^2} x_1^2 + x_2^2 + \lambda(6x_1 + 4x_2 - 9) + \mu(3x_1 + 5x_2 - 7) \\ &= -13\lambda^2 - 8.5\mu^2 - 19\lambda\mu - 9\lambda - 7\mu \end{aligned}$$

Dual problem

$$\begin{aligned} \max \quad & -13\lambda^2 - 8.5\mu^2 - 19\lambda\mu - 9\lambda - 7\mu \\ \text{s.t.} \quad & \lambda \geq 0 \end{aligned}$$

Primal and Lagrange dual problems

- Primal problem

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p \end{array}$$

convex or non-convex

- x is the primal variable, $x \in \mathbf{R}^n$
- optimal value p*

- Lagrange dual problem

$$\max g(\lambda, \nu)$$

subject to: $\lambda \succeq 0$

a convex optimization problem

- λ , ν : dual variables

  $\lambda \in \mathbf{R}^m \quad \nu \in \mathbf{R}^p$

- optimal value d*

Properties of Lagrange dual

Lagrange dual function: $g : \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$,

$$\begin{aligned} g(\lambda, \nu) &= \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \\ &= \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \right) \end{aligned}$$

Property 1: $g(\lambda, \nu)$ is concave on (λ, ν) even if the original problem is not convex

Proof: For each fixed x, $g(\lambda, \nu)$ is linear in (λ, ν). So point-wise infimum of affine (concave) functions of (λ, ν) is concave

Example: the max cut problem

- Consider an undirected graph $G = (V, E)$. For cut $S \subset V$, the capacity of S is the number of edges connecting a node in S to a node not in S. Find $S \subset V$ with maximum capacity.
  - adjacency matrix of the graph

$$Q_{ij} = \begin{cases} 1 & \text{if } (i,j) \in E \\ 0 & \text{otherwise} \end{cases}$$

  - a cut decided by a vector $x \in R^n$

$$x_i = \begin{cases} 1 & \text{if } i \in S \\ -1 & \text{otherwise} \end{cases}$$

  - capacity of the cut

$$c(x) = \frac{1}{4} \sum_{i=1}^{n} \sum_{j=1}^{n} (1 - x_i x_j) Q_{ij}$$

($1 - x_i x_j = 2$ if $\{i, j\}$ is in the cut set)

Example: the max cut problem (cont'd)

- Primal problem (NP complete even if $Q \succeq 0$)

$$\begin{array}{ll} \text{minimize} & x^T Q x \\ \text{subject to} & x_i \in \{-1, 1\} \qquad \text{for all } i = 1, \ldots, n \end{array}$$

the maximum cut is $c_{\max} = \frac{1}{4} \sum_{i=1}^{n} \sum_{j=1}^{n} Q_{ij} - \frac{1}{4} p^\star$

$p^\star$: the optimal value of primal problem

<=>

$$\begin{array}{ll} \text{minimize} & x^T Q x \\ \text{subject to} & x_i^2 - 1 = 0 \end{array}$$

Example: the max cut problem (cont'd)

- Dual problem (SDP)

let $\Lambda = \mathbf{diag}(\lambda_1, \ldots, \lambda_n)$

the Lagrangian is

$$L(x, \lambda) = x^T Q x - \sum_{i=1}^{n} \lambda_i (x_i^2 - 1) = x^T (Q - \Lambda) x + \mathbf{trace}\, \Lambda$$

the dual is

$$\begin{array}{ll} \text{maximize} & \mathbf{trace}\, \Lambda \\ \text{subject to} & Q - \Lambda \succeq 0 \end{array}$$

Properties of Lagrange dual (cont'd)

Property 2: $g(\lambda, \nu) \leq f_0(x)$ for every primal feasible x and dual feasible (λ, ν)

Proof: For primal feasible x, $f_i(x) \leq 0, i = 1, \ldots, m$

$$h_i(x) = 0, i = 1, \ldots, p$$

For dual feasible (λ, ν) $\lambda_i \geq 0, i = 1, \ldots, m$

Then

$$\begin{aligned} f_0(x) &\geq f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \\ &\geq \inf_{x'} (f_0(x') + \sum_{i=1}^{m} \lambda_i f_i(x') + \sum_{i=1}^{p} \nu_i h_i(x')) \\ &= g(\lambda, \nu) \end{aligned}$$

Properties of Lagrange dual (cont'd)

Property 2: $g(\lambda, \nu) \leq f_0(x)$ for every primal feasible x and dual feasible (λ, ν)

Remarks:

(1) $f_0(x) - g(\lambda, \nu)$: duality gap for (x, λ, ν), which is alway non-negative

$p^* - d^*$: optimal duality gap

(2) $d^* \leq p^*$: weak duality

for any optimization problem: convex or not

can be used to find a non-trivial lower bound on p*, for difficult primal problems

Example: the max cut problem (cont'd)

- Any dual feasible solution gives a lower bound on primal objective value:

  for any feasible x

$$x^T Q x \geq x^T \Lambda x = \sum_{i=1}^{n} \Lambda_{ii} x_i^2 = \mathbf{trace}\, \Lambda$$

Strong duality

strong duality: $d^\star = p^\star$

- does not hold in general
- (usually) holds for convex problems
- conditions that guarantee strong duality in convex problems are called **constraint qualifications**

there exist many types of constraint qualifications

Slater's constraint qualification is a famous one

Slater's constraint qualification

strong duality holds for a convex problem

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

if it is strictly feasible, *i.e.*,

$$\exists x \in \mathbf{int}\, \mathcal{D}: \qquad f_i(x) < 0, \quad i = 1, \ldots, m, \qquad Ax = b$$

Can be refined: linear inequalities do not need to hold with strict inequality

Strong duality theorem for LP

- If an LP (in any form) has an optimal solution x^*, then the dual also has an optimal solution y^*, and $C^T x^* = b^T y^*$

Primal

$$\min c^T x$$

subject to:

$$Ax = b$$
$$x \succeq 0$$

Dual

$$\max b^T y$$

subject to:

$$A^T y \preceq c$$

	Primal			
Dual		has optimum	unbounded	infeasible
	has optimum	Y	N	N
	unbounded	N	N	Y
	infeasible	N	Y	Y

- Reference
  - Chapter 5.1 - 5.2, Convex Optimization.
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